REVA UNIVERSITY
Bengaluru, India

TankShift

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Problem Statement

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Pumps are blind to the clock

Campus pumps run on float switches, often in the 6–9 am and 6–10 pm peaks.

Time-of-Day tariff is already live

Karnataka HT consumers ≥ 500 kVA: + ₹1/unit at peak, −₹1/unit at night (Dec–Jun).

Demand charges and the grid

HT billing demand is the month's peak kVA at ₹350–380 per kVA — a pump that starts on the site's peak adds to the bill all month. Facility teams have no cheap tool to act on either. The grid pays too: evening peaks run on the dirtiest plants.

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Solution

TankShift — the tank is the battery

A low-cost retrofit controller on the existing pump starter. It fills the tanks in the cheap night window, locks out the surcharge windows, keeps a hard-coded water reserve, and proves the saving on the ToD bill.

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Solution Design & USP

Sense

Floats and ultrasonic in sump and overhead tank. 3-phase Modbus meter on the pump

Decide

ESP32 holds the local ToD table and site demand profile. Priority: reserve → rebate

Act

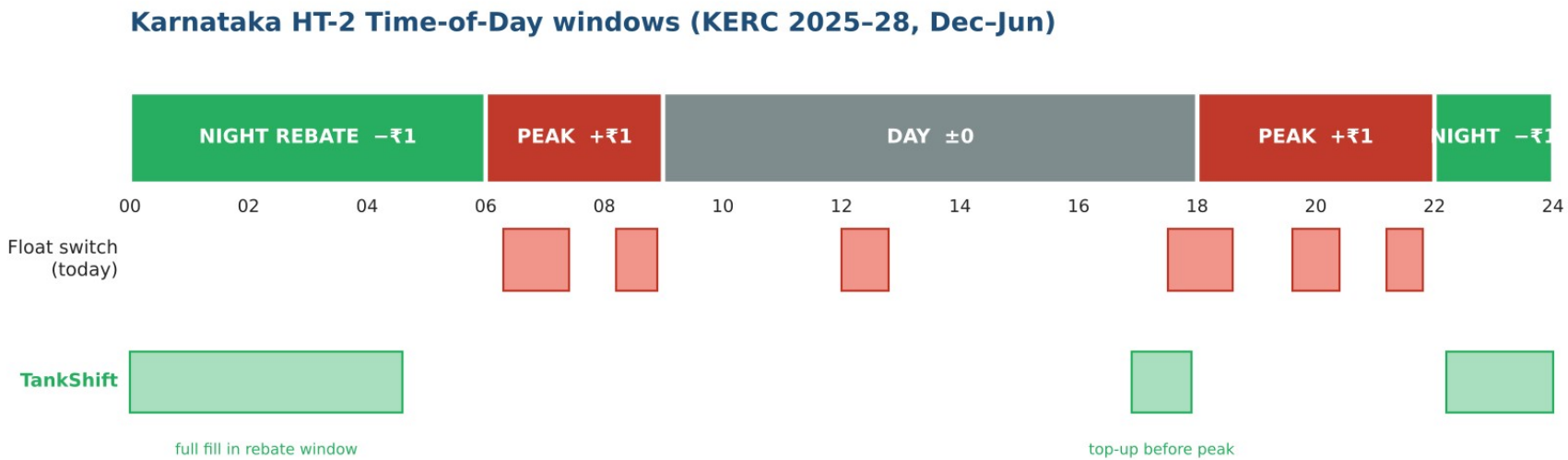
Relay in series with the starter's Auto circuit; manual override still works. USP: timers don't

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Visual of Prototype

Same pump, same water — moved to the hours that cost ₹2 less per unit and keep the grid off its evening peak. Dashboard shows units moved, rupees saved and kVA kept off the site peak.

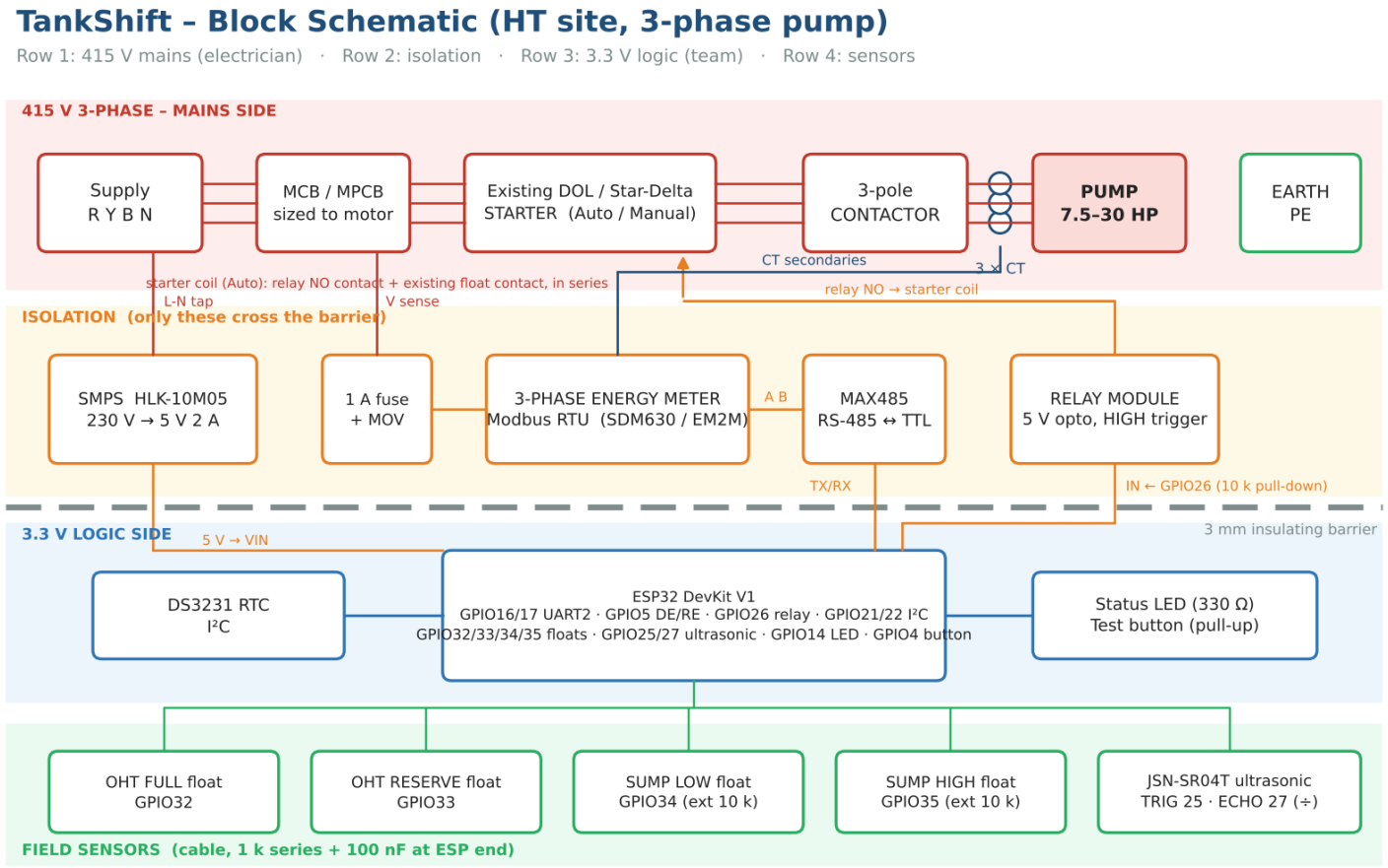


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Circuit Design

Four rows: 415 V mains (electrician), isolation, 3.3 V logic (team), field sensors. The relay only drives the starter coil; the watchman's manual bypass stays.

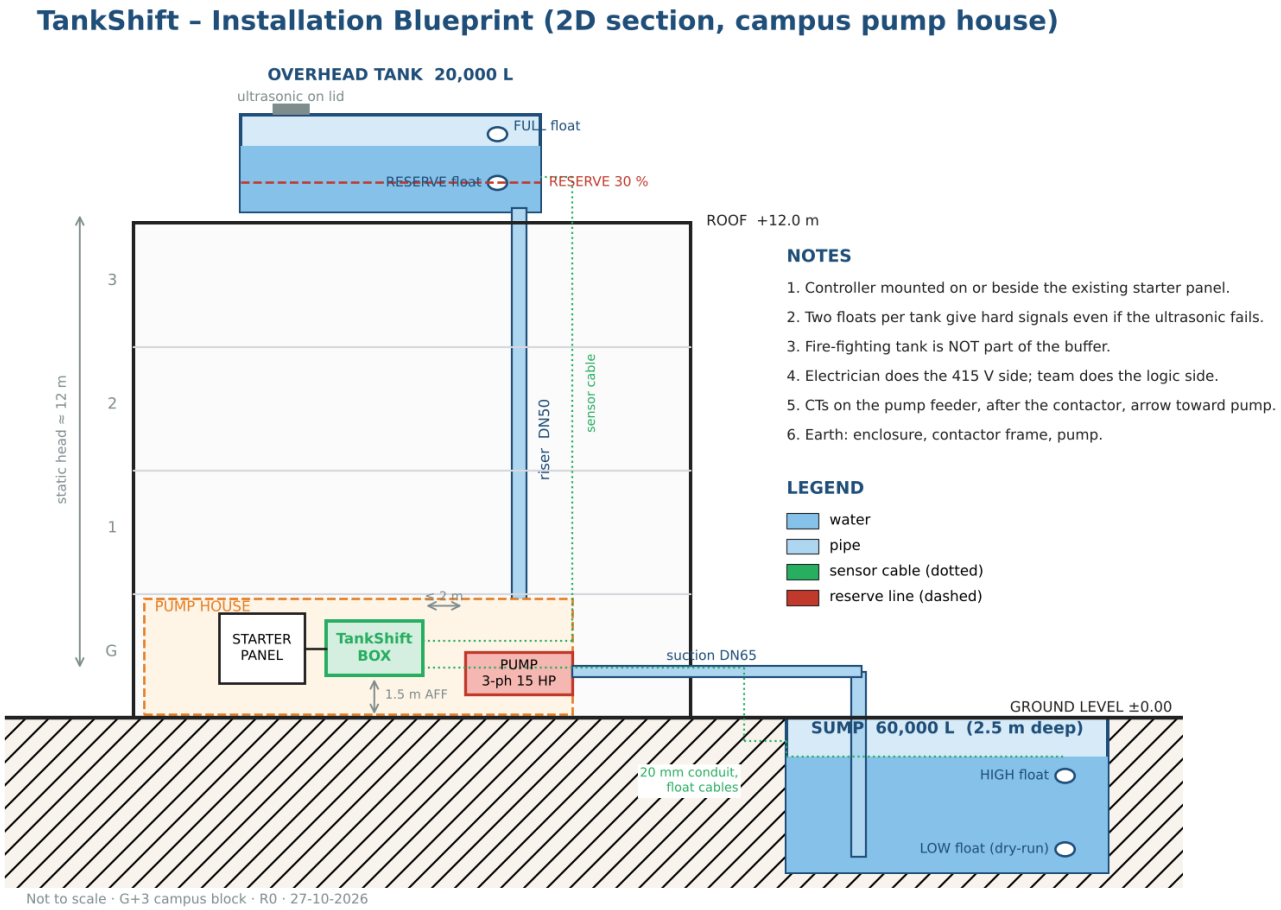


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Site Blueprint

Controller on the existing starter panel in the pump house. Two floats per tank, ultrasonic on the lid, CTs on the pump feeder. Fire tanks are never part of the buffer.



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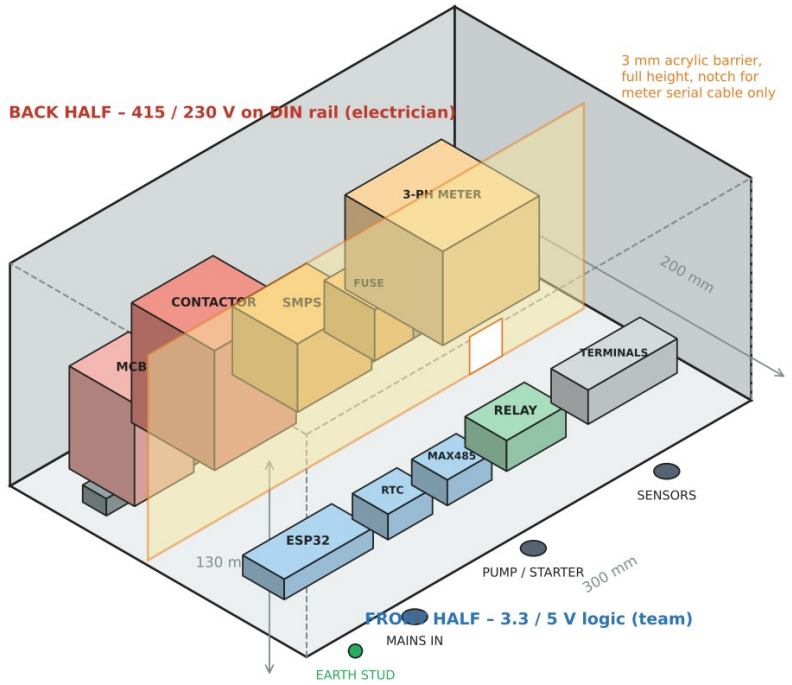
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Enclosure Design

IP65 box, 300 × 200 × 130 mm. Mains on a DIN rail in the back half, logic in the front half, 3 mm barrier between — only the meter cable crosses.

TankShift – Enclosure Cutaway (3D isometric, lid and near walls removed)

IP65 ABS 300 × 200 × 130 mm · mains and logic in separate halves · only the meter serial cable crosses the barrier



BUILD ORDER 1 DIN rail on back wall; MCB, contactor, SMPS, fuse, meter left → right 2 Fit barrier 3 ESP32, RTC, MAX485, relay, terminals in front half 4 Three glands on bottom face + earth stud 5 Label every wire at both ends, then close lid

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Potential Market Size

Every HT-2 consumer ≥ 500 kVA in Karnataka is a compulsory ToD customer: universities, hospitals, IT parks, factories, hotels, malls. 20+ states have ToD for HT; Maharashtra already pays solar-hour rebates.

Campus / year	₹3–5 lakh
Tech park / year	₹15 lakh+
BWSSB / month	₹75 crore

Who pays: the facility (hardware + annual licence), later the DISCOM (demand-response fee). Residential joins when domestic ToD lands — central deadline April 2028.

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Competitors

Timers

Float switches and timers, ₹500–2,000. Blind to tariff, levels and demand.

BMS vendors

Schneider, Honeywell, Siemens: ₹10–50 lakh, only where an energy manager already exists.

Home batteries

Luminous, Tata, Exide: ₹1–5 lakh and they degrade. Water is free.

Energy auditors

A report, not a device. We are the ₹10,000 BMS feature for the sites nobody serves.

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Use of the ₹50,000 Grant

01 Prove it

₹36,000 — 3 pump sets on REVA campus + 2 at a partner institution, 90 days, measured on the ToD bill.

02 Scale it

₹14,000 — 2 more sites + facility dashboard. Released only if Step 1 shows savings on the bill.

03 Report

Pilot report with measured savings from REVA's own HT bill, ready for BESCOM. Unused money returned.

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Conclusion

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Ask: ₹36,000 to run it on REVA's own meter

The grid's evening peak is India's most expensive hour. Campuses have thousands of units of flexible pumping and don't know it. TankShift finds it, moves it, and proves it on the bill.

Thank You